| University of Southampton | School of Electronics and Computer Science | Coursework Instructions |
|---|---|---|
| Module: ELEC6242 | Cryptography | Lecturer: Dr Basel Halak |
| Deadline: 04/03/2026 at 16:00 | Feedback: 04/04/2026 | Weighting: 20% |

## Instructions

This coursework is in three parts. You should complete each part, working independently.

1. **Decrypt the following cipher**

   *You will receive your unique cipher by email*

2. **Decrypt the hidden message contained in the hex file**

   *You will receive a link to your unique cipher by email*

3. **Decrypt the following cipher**

   *You will receive your unique cipher by email*

## File Submission

Please submit your report in **PDF format** using the ECS electronic hand-in system. Submissions in any other formats will not be considered valid submissions, and therefore they may lead to the application of late penalties (10% deduction of the awarded mark for each working day).

## Learning Outcomes (LOs)

Having completed the module, you should be able to:

1. Apply several cryptanalysis methods to decipher encrypted messages
2. Write a concise report describing your proposed solutions, methods of analysis design, and how they were arrived at and performed.

## Marking Scheme

| Criterion | Description | Marks |
|---|---|---|
| *Q1 Answer* | *The accuracy of your decryption* | *10* |
| *Q1 Methodology* | *The robustness of your technique in terms of the quality of your evidence and the depth of your analysis* | *30* |
| *Q2 Answer* | *The accuracy of your decryption* | *5* |
| *Q2 Methodology* | *The robustness of your technique in terms of the quality of your evidence and the depth of your analysis* | *20* |
| *Q3 Answer* | *The accuracy of your decryption* | *5* |
| *Q3 Methodology* | *The robustness of your technique in terms of the quality of your evidence and the depth of your analysis* | *20* |
| *Report Writing* | *The quality of your report in terms of its clarity and organization* | *10* |

## Report Structure

**You will need to use the same structure provided in the answer sheet.**

**Please note failure to follow this structure will lead to a loss of marks.**

Your report should have the following structure:

1. **Solution for Cipher 1**: this section only includes the plaintext for cipher 1 and the decryption key

2. **Solution for Cipher 2**: this section only includes the plaintext for cipher 2 and the decryption key
3. **Solution for Cipher 3**: this section only includes the plaintext for cipher 3 and the decryption key
4. **Cipher 1 Cryptanalysis**: This section provides a summary of the techniques you have used to solve the cipher. It should include evidence of all analysis you have performed to solve the cipher such (e.g. frequency analysis), in addition, it should refer to any programs you have developed to perform cryptanalysis
5. **Cipher 2 Cryptanalysis**: This section provides a summary of the techniques you have used to solve the cipher. It should include evidence of all analysis you have performed to solve the cipher, in addition, it should refer to any programs you have developed to perform cryptanalysis
6. **Cipher 3 Cryptanalysis**: This section provides a summary of the techniques you have used to solve the cipher. It should include evidence of all analysis you have performed to solve the cipher in addition it should refer to any programs you have developed to perform cryptanalysis
7. **Appendix A:** includes any software scripts you have developed to solve cipher 1
8. **Appendix B**: includes any software scripts you have developed to solve cipher 2
9. **Appendix C**: includes any software scripts you have developed to solve cipher 3

*Late submissions will be penalized at 10% per working day. No work can be accepted after feedback has been given.*
*Please note the University regulations regarding academic integrity.*